

Zero-Drift, Bi-Directional Current Sense Amplifier

Features

- Voltage Offset: $\pm 100 \mu\text{V}$ (Max)
- Wide Common-Mode Voltage: -0.3 V to $+36 \text{ V}$
- Supply Voltage: 2.7 V to $+30 \text{ V}$
- Accuracy and Zero-Drift Performance
 - $\pm 1\%$ Gain Error (Max over Temperature)
 - $0.5\text{-}\mu\text{V}/^\circ\text{C}$ Offset Drift (Max)
 - $10\text{-ppm}/^\circ\text{C}$ Gain Drift (Max)
- Three Gain Options for Voltage Output
 - TP181A1: 50 V/V
 - TP181A2: 100 V/V
 - TP181A3: 200 V/V
- Low Supply Current: $120 \mu\text{A}$ (Typ)
- Rail-to-Rail Output
- Package: SOT363 (SC70-6)
- Industrial Operation Range: -40°C to 125°C
- ESD Rating: Robust 2-kV HBM, 2-kV CDM
- Higher Performance Drop-in Compatible with INA213, INA214, INA199, NCS199 Products

Description

The TP181 is a series of zero-drift, bi-directional current sense amplifiers that can sense voltage drops across shunts at common-mode voltages from -0.3 V to 36 V , independent of the supply voltage. Three fixed gains are available: 50 V/V , 100 V/V , and 200 V/V . The low offset of the zero-drift architecture enables the current sensing with maximum drops across the shunt as low as 10 mV full-scale.

The TP181 series operates from a single $+2.7\text{-V}$ to 30-V power supply, drawing a supply current of $120 \mu\text{A}$ typical. The series is specified from -40°C to $+125^\circ\text{C}$, and offered in the SOT363 (SC70-6) package.

Table 1. Gain Options

Product	Gain	R3 and R4	R1 and R2
TP181A1	50	$20 \text{ k}\Omega$	$1 \text{ M}\Omega$
TP181A2	100	$10 \text{ k}\Omega$	$1 \text{ M}\Omega$
TP181A3	200	$5 \text{ k}\Omega$	$1 \text{ M}\Omega$

$$V_{\text{OUT}} = (I_{\text{LOAD}} \times R_{\text{SHUNT}}) \text{ GAIN} + V_{\text{REF}}$$

Applications

- Current Sensing (High-Side/Low-Side)
- Battery Chargers
- Power Management
- Cell Phone Charger
- Electrical Cigarette
- Wireless Charger
- Telecom Equipment

Typical Application Circuit

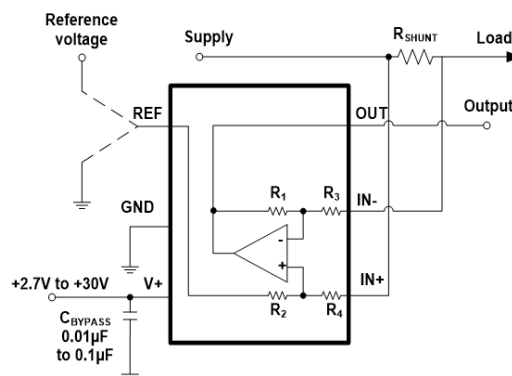


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Revision History

Date	Revision	Notes
2026-04-20	Rev.A.4	The following updates are all about the new datasheet formats or typos, and the actual product remains unchanged. • Updated to a new datasheet format. • Updated the test conditions in the EC table. • Updated the Tape and Reel Information • Updated the POD.

Pin Configuration and Functions

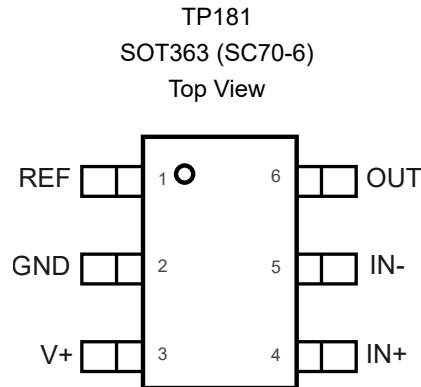


Table 2. Pin Functions: TP181

Pin No.	Name	I/O	Description
1	REF		Reference voltage.
2	GND		Negative power supply.
3	V+		Positive power supply. Typically, the voltage is from 2.7 V to 30 V. A bypass capacitor of 0.1 μ F as close to the part as possible should be used between the power supply pin and GND.
4	IN+	I	Non-inverting input of the amplifier.
5	IN-	I	Inverting input of the amplifier.
6	OUT	O	Output of the amplifier.

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Specifications

Absolute Maximum Ratings ⁽¹⁾

Parameter		Min	Max	Unit
	Supply Voltage		42	V
	Input Voltage	GND – 0.3	42	V
	Input Current: +IN, –IN ⁽²⁾	–10	10	mA
	Output Current: OUT	–35	35	mA
	Current at Supply Pins	–60	60	mA
T _J	Maximum Junction Temperature		150	°C
T _A	Operating Temperature Range	–40	125	°C
T _{STG}	Storage Temperature Range	–65	150	°C
T _L	Lead Temperature (Soldering, 10 sec)		260	°C

(1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Exposure to any Absolute Maximum Rating condition for extended periods may affect device reliability and lifetime.

(2) The input current should be limited to less than 10 mA if the input voltage exceeds the absolute maximum ratings.

ESD, Electrostatic Discharge Protection

Symbol	Parameter	Condition	Minimum Level	Unit
HBM	Human Body Model ESD	ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2	kV
CDM	Charged Device Model ESD	ANSI/ESDA/JEDEC JS-002 ⁽²⁾	±2	kV

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

Thermal Information

Package Type	θ _{JA}	θ _{JC}	Unit
SOT363 (SC70-6)	227	80	°C/W

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Electrical Characteristics

All test conditions: $T_A = 25^\circ\text{C}$, $V_{\text{SENSE}} = V_{\text{IN}+} - V_{\text{IN}-}$, $V_S = 5\text{ V}$, $V_{\text{IN}+} = 12\text{ V}$, $V_{\text{REF}} = V_S / 2$, unless otherwise noted.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Input						
V_{OS}	Input Offset Voltage	$V_{\text{SENSE}} = 0\text{ mV}$ $V_{\text{IN}+} = 0\text{ V}$		± 10	± 100	μV
$V_{\text{OS TC}}$	Input Offset Voltage Drift	$V_{\text{SENSE}} = 0\text{ mV}$, -40°C to 125°C $V_{\text{IN}+} = 0\text{ V}$		0.1	0.5	$\mu\text{V}/^\circ\text{C}$
V_{CM}	Common-Mode Input Range	-40°C to 125°C	-0.3		36	V
CMRR	Common-Mode Rejection Ratio	$V_{\text{IN}+} = 5\text{ V}$ to 26 V , $V_{\text{SENSE}} = 0\text{ mV}$, -40°C to 125°C	95	120		dB
I_B	Input Bias Current	$V_{\text{SENSE}} = 0\text{ mV}$		35		μA
I_{OS}	Input Offset Current	$V_{\text{SENSE}} = 0\text{ mV}$		0.4		μA
PSRR	Power Supply Rejection Ratio	$V_S = 2.7\text{ V}$ to 18 V , $V_{\text{IN}+} = +18\text{ V}$, $V_{\text{SENSE}} = 0\text{ mV}$		± 1		$\mu\text{V}/\text{V}$
Noise R_{ti} ⁽¹⁾						
e_n	Input Voltage Noise Density	$f = 1\text{ kHz}$		30		$\text{nV}/\sqrt{\text{Hz}}$
Output						
G	Gain	TP181A1		50		V/V
		TP181A2		100		V/V
		TP181A3		200		V/V
GE	Gain Error	$V_{\text{SENSE}} = -5$ to 5 mV , -40°C to 125°C		± 0.1	± 1	%
GE TC	Gain Error vs. Temperature	-40°C to 125°C		3	10	ppm
C_{LOAD}	Maximum Capacitive Load	No oscillation		1		nF
V_{OH}	Output Swing from Supply Rail	$R_{\text{LOAD}} = 10\text{ k}\Omega$ to REF, -40°C to 125°C		0.02	0.05	V
V_{OL}	Output Swing from Supply Rail	$R_{\text{LOAD}} = 10\text{ k}\Omega$ to REF, -40°C to 125°C		0.01	0.05	V
Frequency Response						
BW	Bandwidth	$C_{\text{LOAD}} = 10\text{ pF}$, TP181A1		48		kHz
		$C_{\text{LOAD}} = 10\text{ pF}$, TP181A2		30		kHz
		$C_{\text{LOAD}} = 10\text{ pF}$, TP181A3		20		kHz
SR	Slew Rate			0.6		$\text{V}/\mu\text{s}$
Power Supply						

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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V+	Supply Voltage		2.7		30	V
I _Q	Quiescent Current	V _{SENSE} = 0 mV		120	150	μA
Temperature Range						
Specified Range			-40		125	°C
Operating Range			-55		150	°C

(1) RTI = referred to input.

Typical Performance Characteristics

All test conditions: $T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{REF} = V_S / 2$, unless otherwise noted.

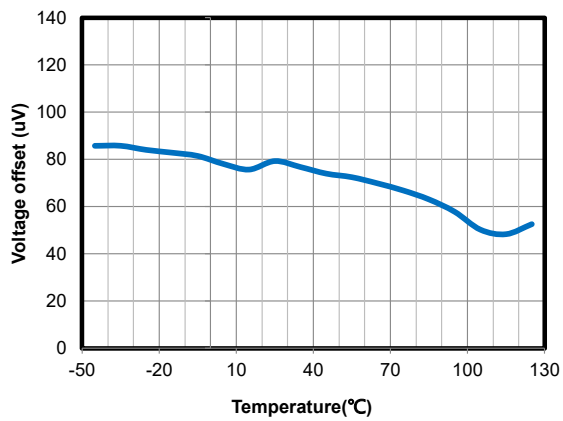


Figure 1

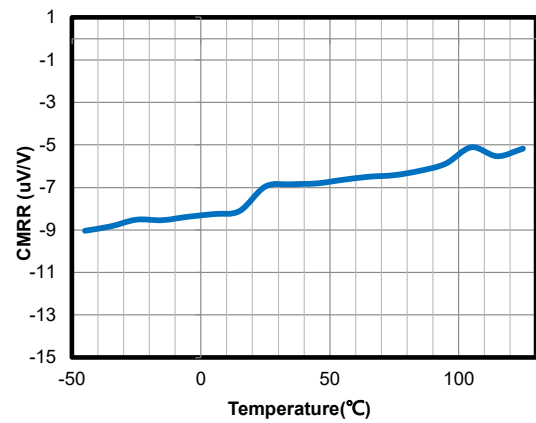


Figure 2

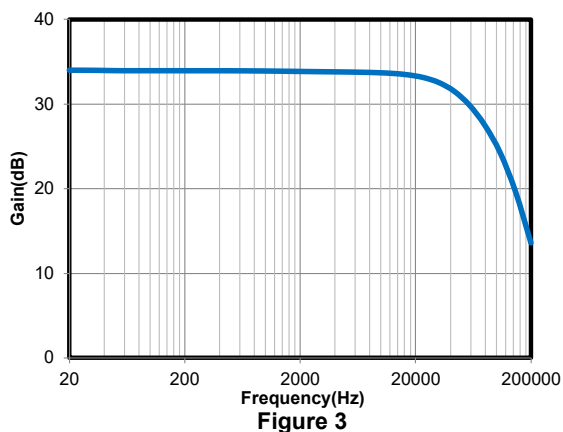


Figure 3

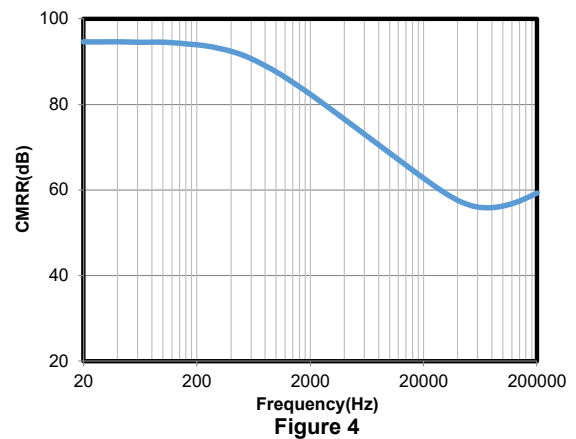
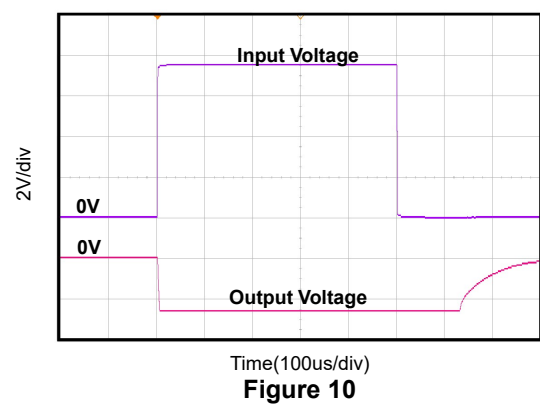
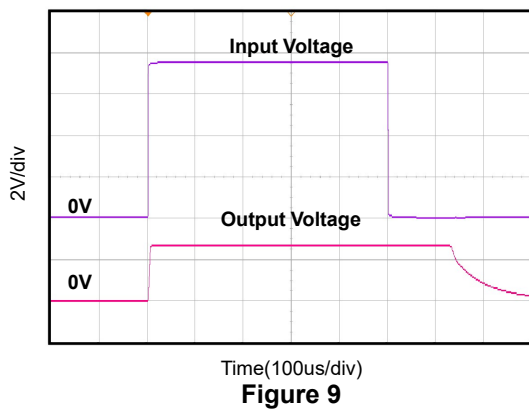
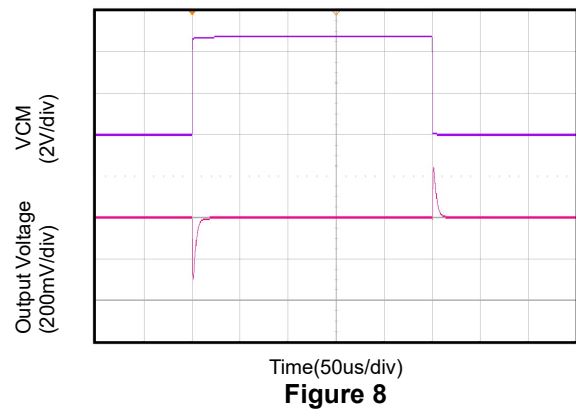
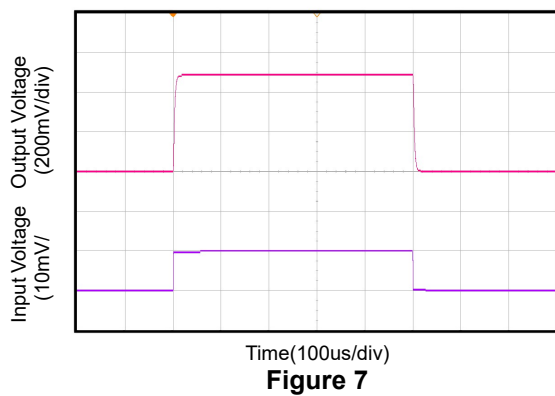
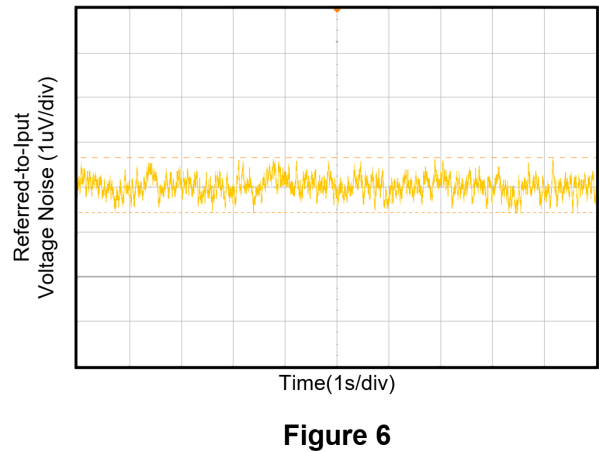
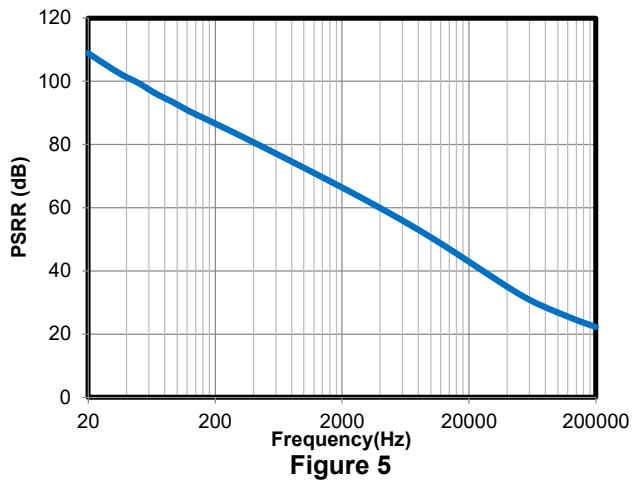
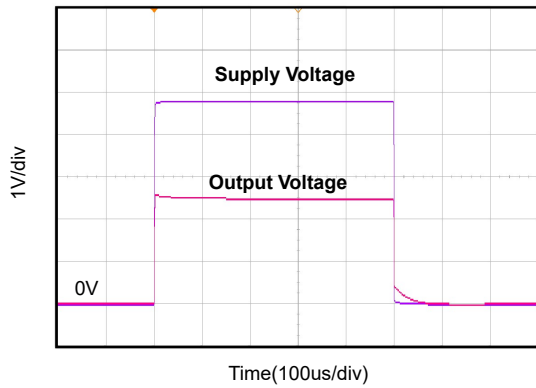
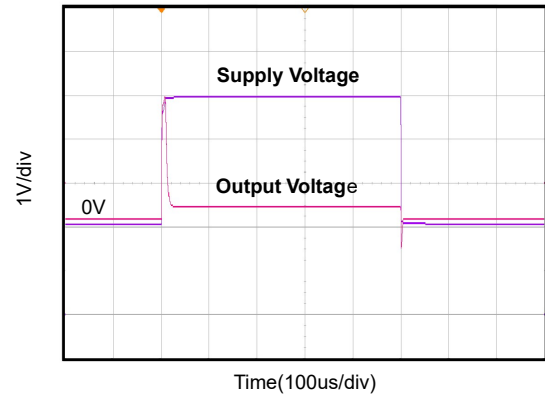
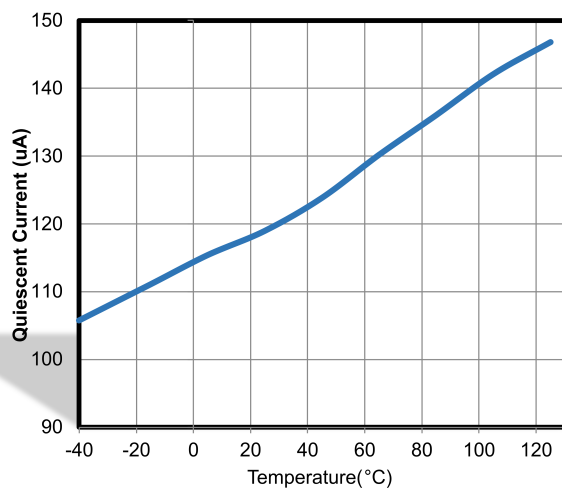
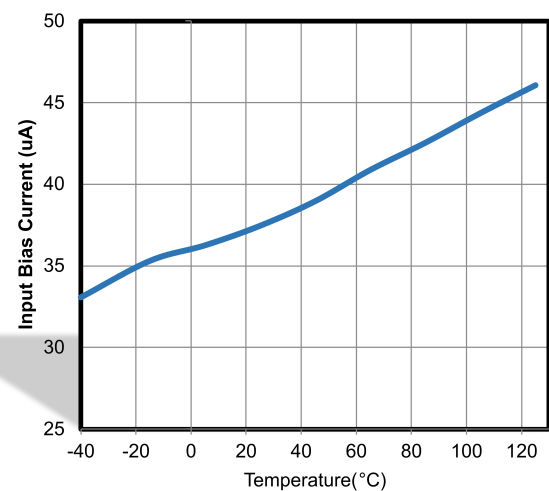
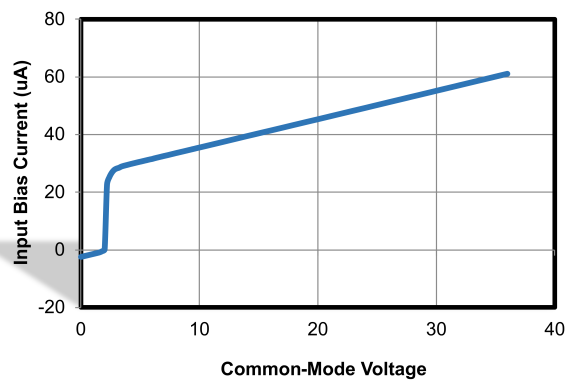


Figure 4

Zero-Drift, Bi-Directional Current Sense Amplifier



Zero-Drift, Bi-Directional Current Sense Amplifier


Figure 11

Figure 12

Figure 13

Figure 14

Figure 15

Zero-Drift, Bi-Directional Current Sense Amplifier

Detailed Description

Overview

The TP181 is a series of 36-V common-mode, zero-drift topology, current-sensing amplifiers that can be used in both low-side and high-side configurations. The specially-designed, current-sensing amplifiers are able to accurately measure voltages developed across current-sensing resistors on common-mode voltages that far exceed the supply voltage powering the device. The current can be measured on input voltage rails as high as 36 V while the device can be powered from supply voltages as low as 2.7 V.

The zero-drift topology enables high-precision measurements with maximum input offset voltages as low as 100 μV with a maximum temperature contribution of 0.5 $\mu\text{V}/^{\circ}\text{C}$ over the full temperature range from -40°C to 125°C .

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Application and Implementation

Note

Information in the following application sections is not part of the 3PEAK's component specification and 3PEAK does not warrant its accuracy or completeness. 3PEAK's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

Application Information

Application Schematic

Figure 16 shows the basic connections of the TP181. The input pins, IN+ and IN–, should be connected as close as possible to the shunt resistor to minimize any resistance in series with the shunt resistor.

Power-supply bypass capacitors are required for stability. Applications with noisy or high-impedance power supplies may require additional decoupling capacitors to reject power-supply noise. Connect bypass capacitors close to the pins of the device.

Selecting R_{SHUNT}

The zero-drift offset performance of the TP181 offers several benefits. Most often, the primary advantage of the low offset characteristic enables lower full-scale drops across the shunt. For example, non-zero-drift current shunt monitors typically require a full-scale range of 100 mV.

The TP181 series gives equivalent accuracy at a full-scale range on the order of 10 mV. This accuracy reduces shunt dissipation by an order of magnitude with many additional benefits.

Alternatively, there are applications that must measure current over a wide dynamic range that can use the low offset on the low end of the measurement. Most often, these applications can use the lower gains of the TP181 to accommodate larger shunt drops on the upper end of the scale. For instance, the TP181A1 operating on a 3.3-V supply can easily handle a full-scale shunt drop of 60 mV, with only 100- μ V of offset.

REF Input Impedance Effects

As with any difference amplifier, the TP181 series common-mode rejection ratio is affected by any impedance present at the REF input. This concern is not a problem when the REF pin is connected directly to most references or power supplies. When using resistive dividers from the power supply or a reference voltage, the REF pin should be buffered by an op-amp.

Power Supply Recommendation

The input circuitry of the TP181 can accurately measure beyond its power-supply voltage, V+. For example, the V+ power supply can be 5 V, whereas the load power-supply voltage can be as high as 30 V. However, the output voltage range of the OUT pin is limited by the voltages on the power-supply pin. Note also that the TP181 can withstand the full input signal range up to 36 V at the input pins, regardless of whether the device has power applied or not.

Proper Board Layout

To ensure optimum performance at the PCB level, care must be taken in the design of the board layout. To avoid leakage currents, the surface of the board should be kept clean and free of moisture. The coating of the surface creates a barrier to the moisture accumulation and helps reduce the parasitic resistance on the board.

Keeping supply traces short and properly bypassing the power supplies can minimize power supply disturbances due to the output current variation, for example when driving an AC signal into a heavy load. Bypass capacitors should be connected

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as closely as possible to the supply pins of the device. The stray capacitance is a concern at the inputs and outputs of the amplifier. It is recommended that signal traces should be kept at least 5 mm from supply lines to minimize the coupling.

A variation in temperature across the PCB can cause a mismatch in the Seebeck voltages at solder joints and other points where dissimilar metals are in contact, resulting in thermal voltage errors. To minimize these thermocouple effects, orient the resistors, so heat sources can warm both ends equally. Input signal paths should contain matching numbers and types of components, where possible to match the numbers and types of thermocouple junctions. For example, dummy components such as zero-value resistors can be used to match real resistors in the opposite input path. Matching components should be in close proximity and oriented in the same manner. Ensure the leads are of equal length so that the thermal conduction is in equilibrium. Keep heat sources on the PCB as far away from the amplifier input circuitry as is practical.

The use of a ground plane is highly recommended. A ground plane reduces the EMI noise and helps to maintain a constant temperature across the circuit board.

Typical Application

Figure 16 shows the typical application schematic.

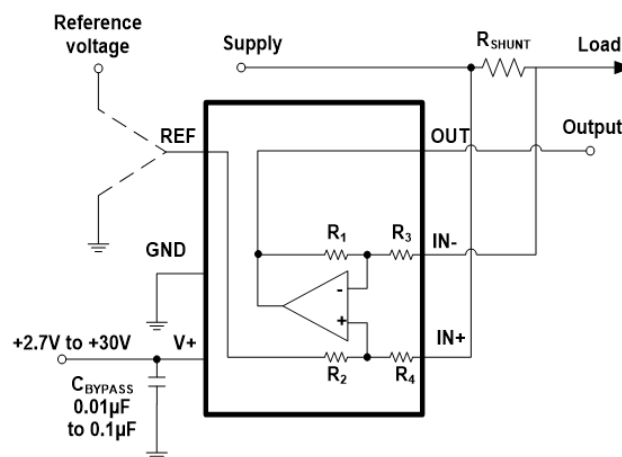
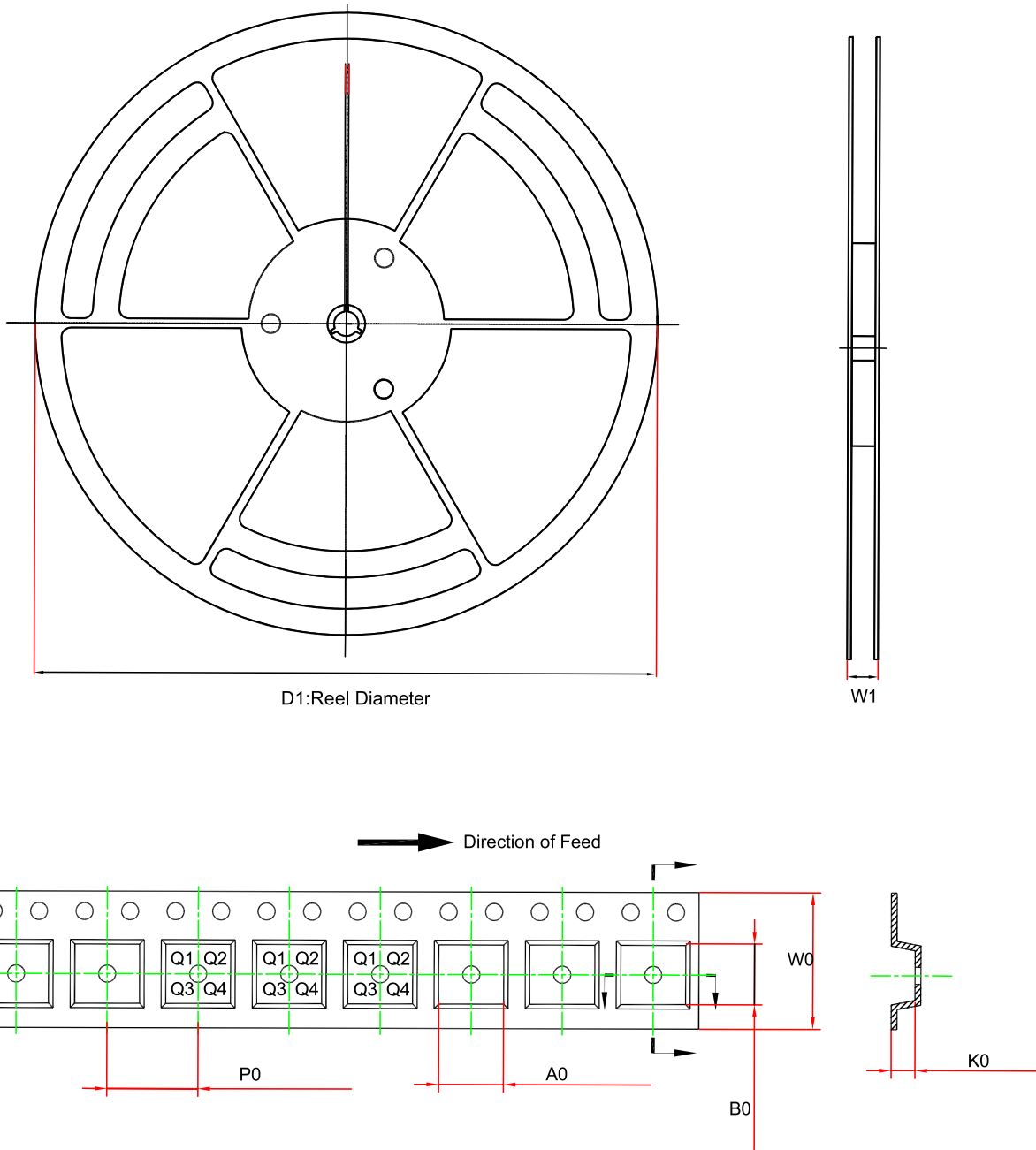


Figure 16. Typical Application Circuit

Tape and Reel Information



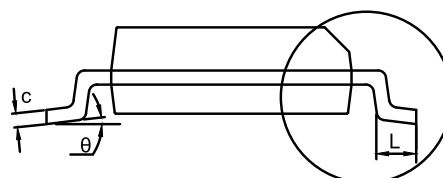
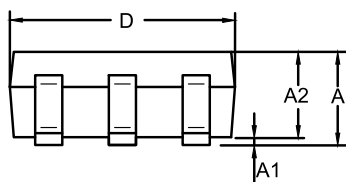
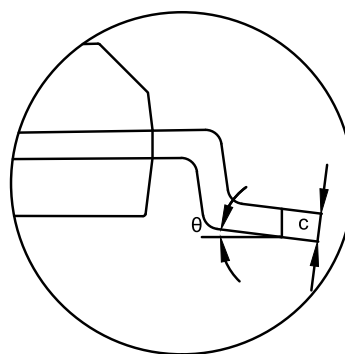
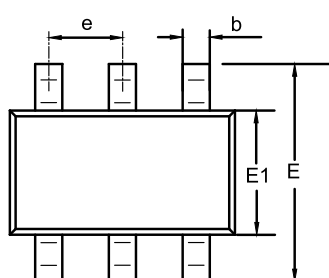
Order Number	Package	D1 (mm)	W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P0 (mm)	W0 (mm)	Pin1 Quadrant
TP181x-CR	SOT-363	178.0	12.1	2.4	2.5	1.2	4.0	8.0	Q3

Package Outline Dimensions

SOT363

Package Outline Dimensions

SC6(SOT363-6-A)



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	MIN	MAX	MIN	MAX
A	0.850	1.100	0.033	0.043
A1	0.000	0.100	0.000	0.004
A2	0.800	1.000	0.031	0.039
b	0.150	0.350	0.006	0.014
c	0.080	0.230	0.003	0.009
D	2.000	2.200	0.079	0.087
E	2.150	2.450	0.085	0.096
E1	1.150	1.350	0.045	0.053
e	0.650 BSC		0.026 BSC	
L	0.260	0.460	0.010	0.018
θ	0	8°	0	8°

NOTES

1. Do not include mold flash or protrusion.
2. This drawing is subject to change without notice.

Order Information

Order Number	Gain	Package	Marking Information	MSL	Transport Media, Quantity	Eco Plan
TP181A1-CR	50 V/V	SOT363 (SC70-6)	9A1	1	Tape and Reel, 3,000	Green
TP181A2-CR	100 V/V	SOT363 (SC70-6)	9A2	1	Tape and Reel, 3,000	Green
TP181A3-CR	200 V/V	SOT363 (SC70-6)	9A3	1	Tape and Reel, 3,000	Green

Green: 3PEAK defines "Green" to mean RoHS compatible and free of halogen substances.

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